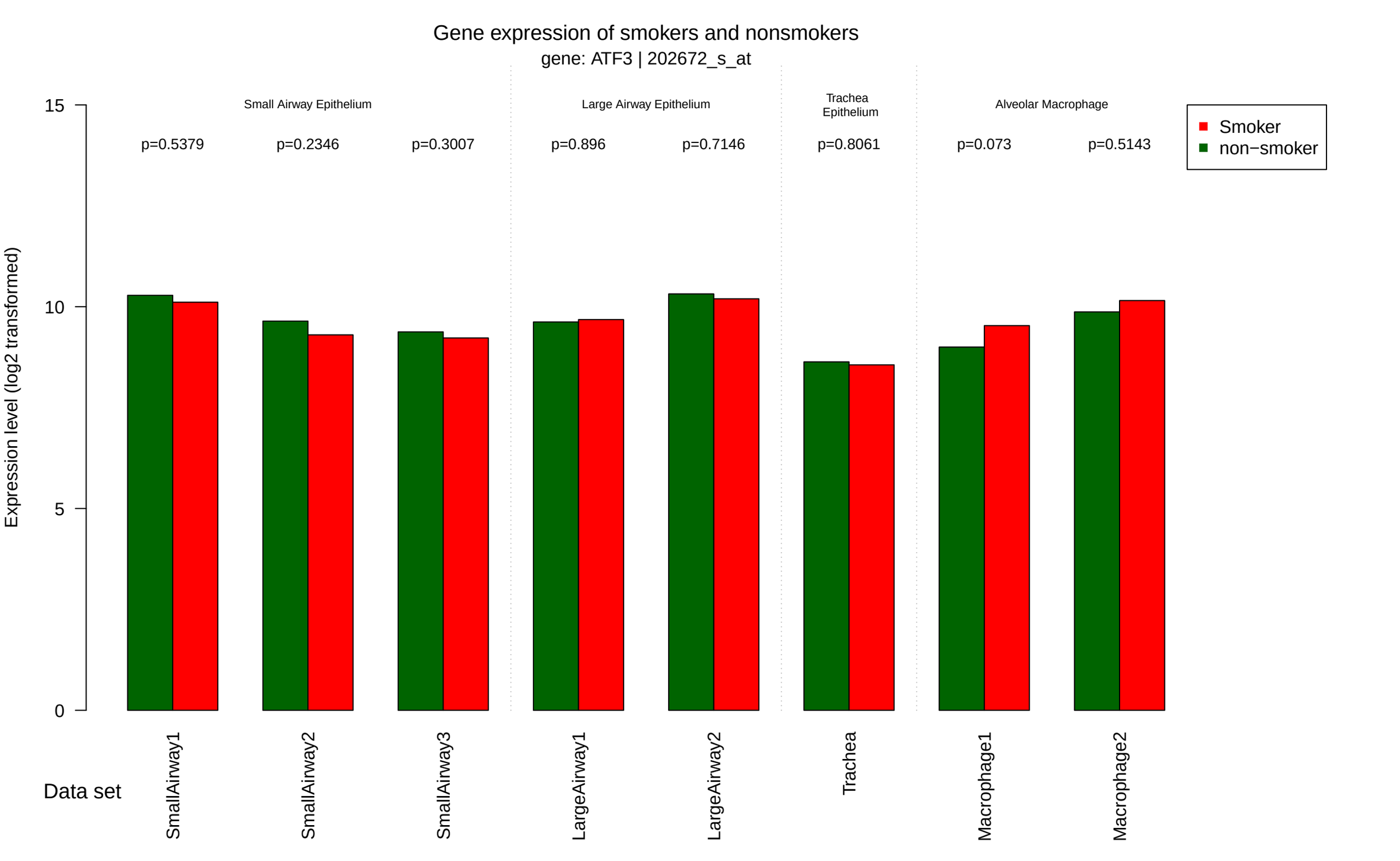
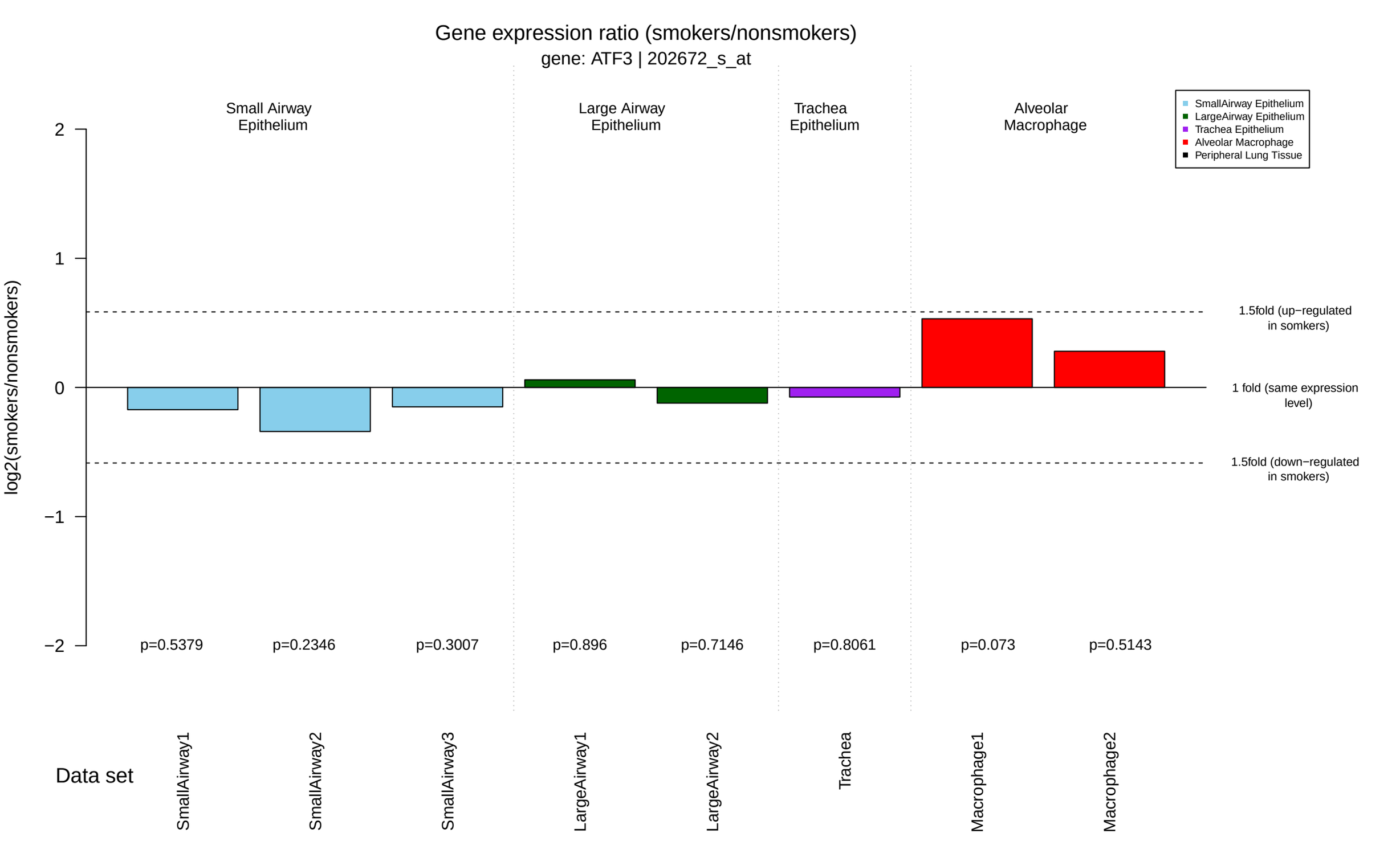
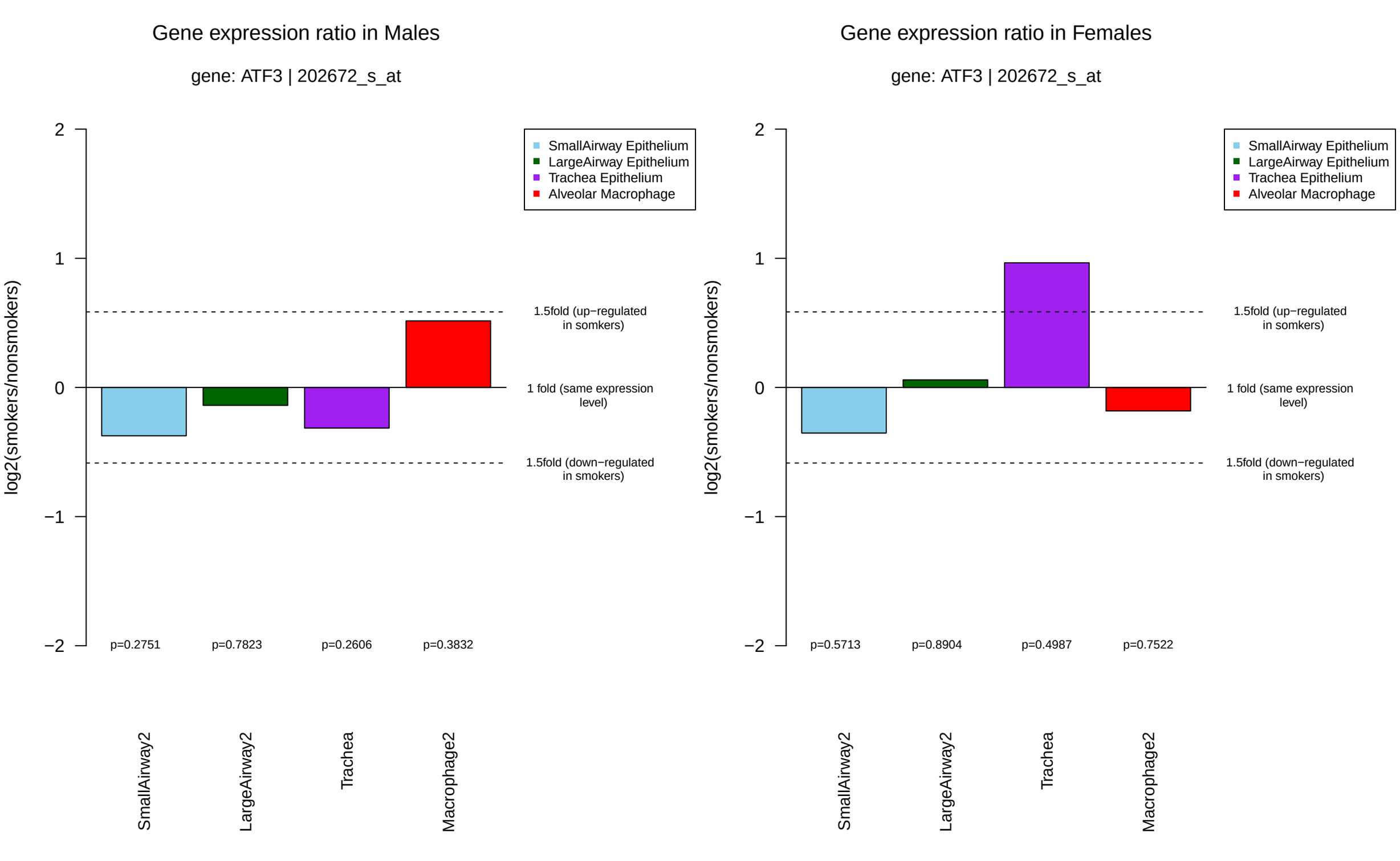
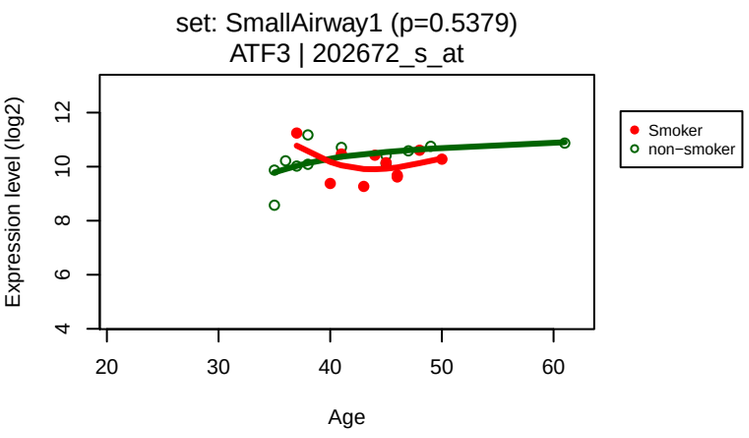
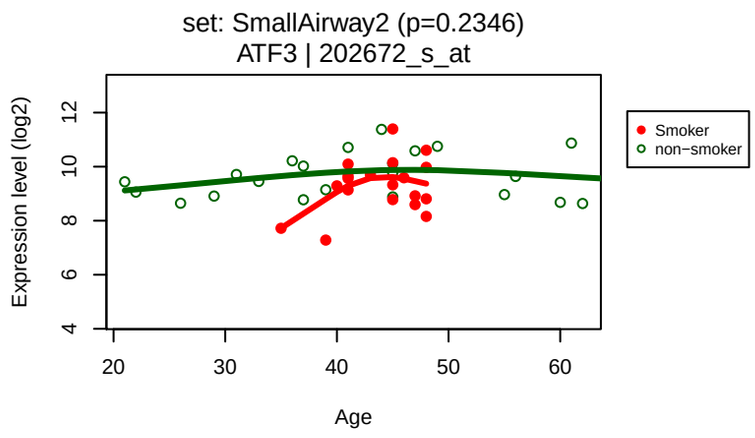
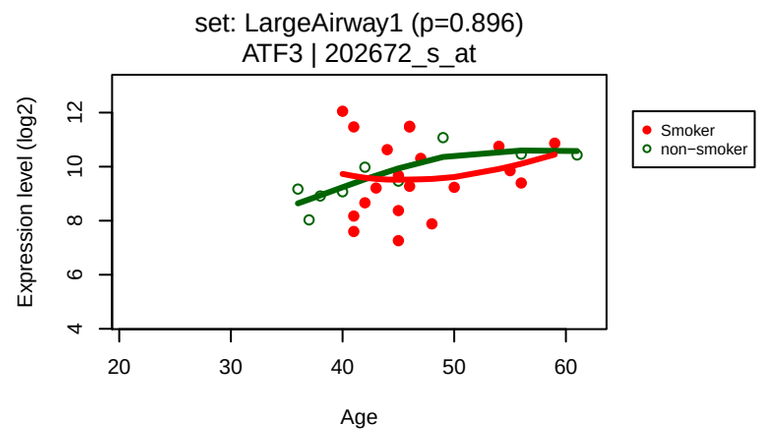

Gene expression of smokers and nonsmokers
gene: ATF3 | 202672_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5379
p=0.2346
p=0.3007
p=0.896
p=0.7146
p=0.8061
p=0.073
p=0.5143
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

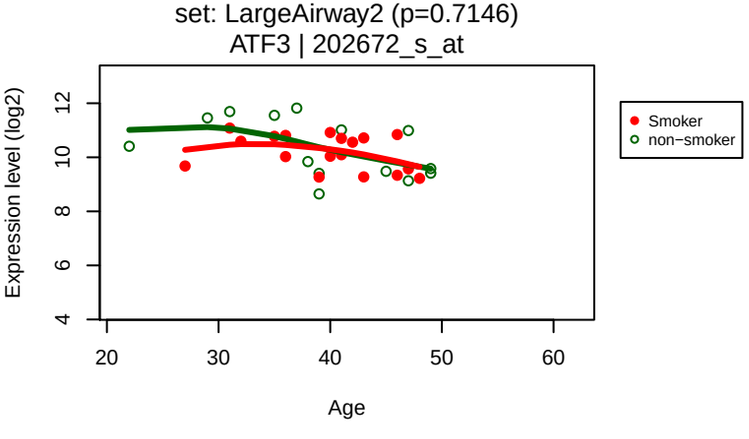

Gene expression ratio (smokers/nonsmokers)
gene: ATF3 | 202672_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5379
p=0.2346
p=0.3007
p=0.896
p=0.7146
p=0.8061
p=0.073
p=0.5143
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

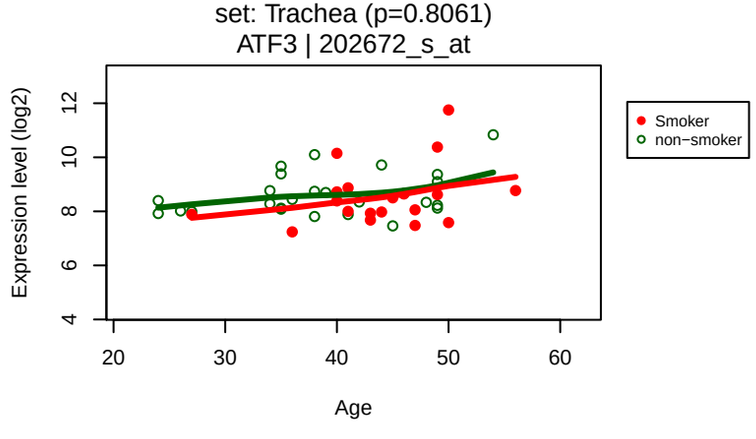

Gene expression ratio in Males
gene: ATF3 | 202672_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2751
p=0.7823
p=0.2606
p=0.3832
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF3 | 202672_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5713
p=0.8904
p=0.4987
p=0.7522
SmallAirway2
LargeAirway2
Trachea
Macrophage2

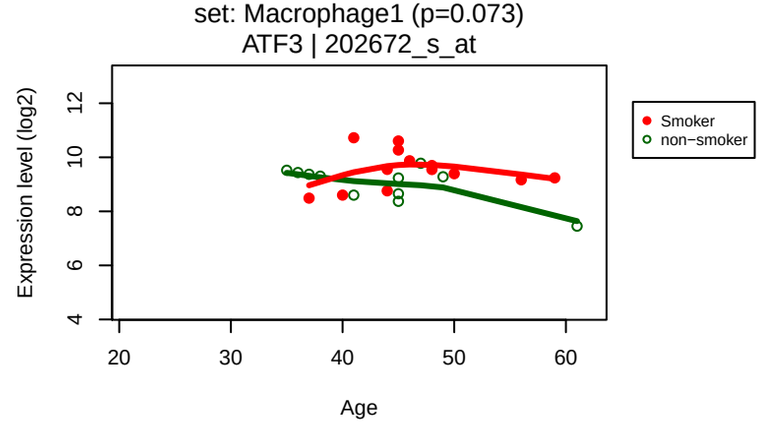

set: SmallAirway1 (p=0.5379)
ATF3 | 202672_s_at
Expression level (log2)
Age
Smoker
non−smoker

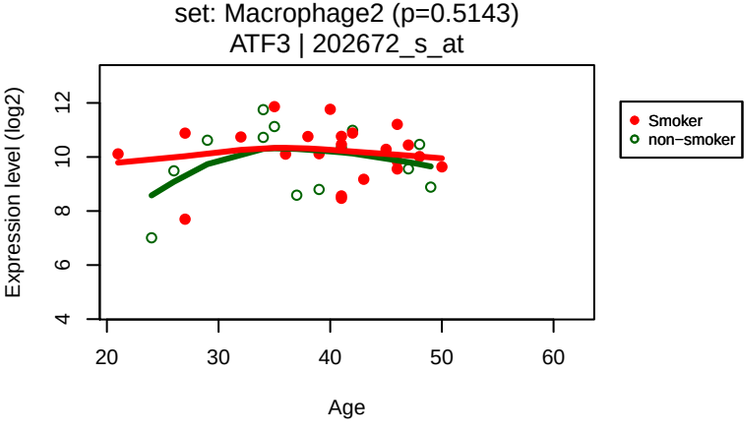

set: SmallAirway2 (p=0.2346)
ATF3 | 202672_s_at
Expression level (log2)
Age
Smoker
non−smoker

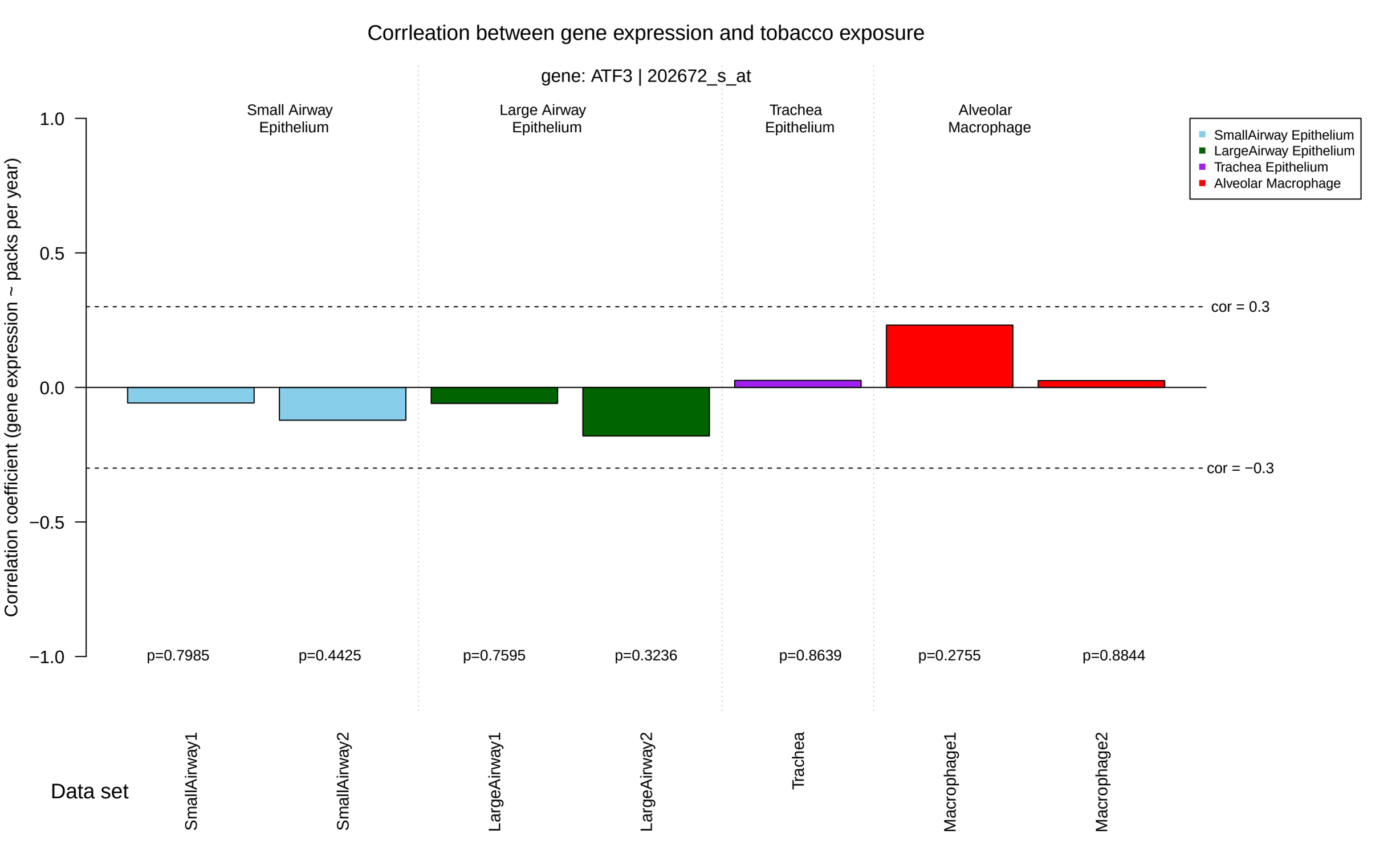

set: LargeAirway1 (p=0.896)
ATF3 | 202672_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7146)
ATF3 | 202672_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8061)
ATF3 | 202672_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.073)
ATF3 | 202672_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5143)
ATF3 | 202672_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF3 | 202672_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7985
p=0.4425
p=0.7595
p=0.3236
p=0.8639
p=0.2755
p=0.8844
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2